BIOL 343
Applied Bioinformatics

DEG analysis

Dr. Nic Wheeler

Counting converts mapped reads to expression values

Expression values will be used by statistical models to identify DEGs

- Recall the goal of our RNA-seq experiments…
  - Treatment vs Control
  - Mutant vs Wild type
  - ***Identify differentially expressed genes (DEGs)***
- DEGs will be identified using statistical tests comparing ***expression values*** of transcripts/genes
- Expression values will be calculated based on the number of reads that ***align/map*** to a given genomic locus
- ***So you have DEGs…now what?***

DEGs are analyzed and evaluated by content experts

Sometimes they are a bioinformatician, sometimes not

DEGs are analyzed and evaluated by content experts

Sometimes they are a bioinformatician, sometimes not

- DEGs are list of genes that ***may*** be important in the response to a perturbation (treatment, mutation, etc.)

DEGs are analyzed and evaluated by content experts

Sometimes they are a bioinformatician, sometimes not

- DEGs are list of genes that ***may*** be important in the response to a perturbation (treatment, mutation, etc.)
- Just because they are differentially expressed does not mean they are biologically meaningful in that context

DEGs are analyzed and evaluated by content experts

Sometimes they are a bioinformatician, sometimes not

- DEGs are list of genes that ***may*** be important in the response to a perturbation (treatment, mutation, etc.)
- Just because they are differentially expressed does not mean they are biologically meaningful in that context
- Lists of DEGs are provided to the subject matter experts - the people who designed the experiment

DEGs are analyzed and evaluated by content experts

Sometimes they are a bioinformatician, sometimes not

- DEGs are list of genes that ***may*** be important in the response to a perturbation (treatment, mutation, etc.)
- Just because they are differentially expressed does not mean they are biologically meaningful in that context
- Lists of DEGs are provided to the subject matter experts - the people who designed the experiment
- Subject matter experts explore DEGs to build new hypotheses

DEGs are analyzed and evaluated by content experts

Sometimes they are a bioinformatician, sometimes not

- DEGs are list of genes that ***may*** be important in the response to a perturbation (treatment, mutation, etc.)
- Just because they are differentially expressed does not mean they are biologically meaningful in that context
- Lists of DEGs are provided to the subject matter experts - the people who designed the experiment
- Subject matter experts explore DEGs to build new hypotheses
- DEGs hypothesized to be responsive to the perturbation need to be followed up with new experiments…otherwise it is just speculation

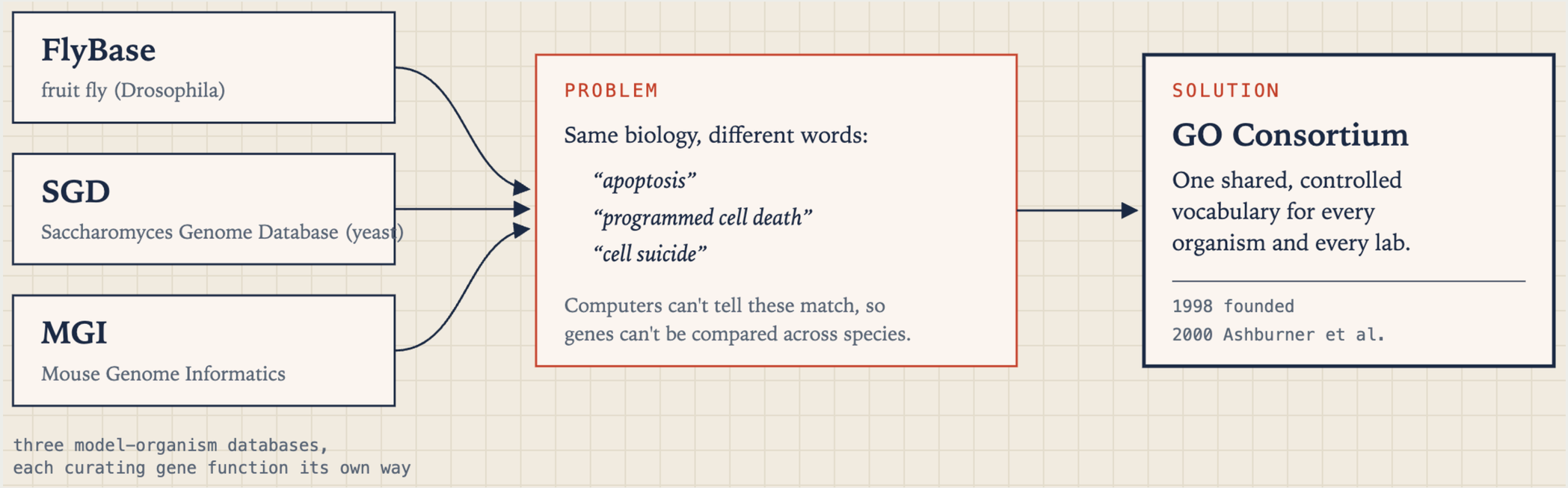
DEGs are analyzed and evaluated by content experts

Sometimes they are a bioinformatician, sometimes not

- DEGs are list of genes that ***may*** be important in the response to a perturbation (treatment, mutation, etc.)
- Just because they are differentially expressed does not mean they are biologically meaningful in that context
- Lists of DEGs are provided to the subject matter experts - the people who designed the experiment
- Subject matter experts explore DEGs to build new hypotheses
- DEGs hypothesized to be responsive to the perturbation need to be followed up with new experiments…otherwise it is just speculation
- There are a few ways that almost all RNA-seq analyses will begin to explore DEGs to look for biological relevance…

Gene Ontology (GO) annotations

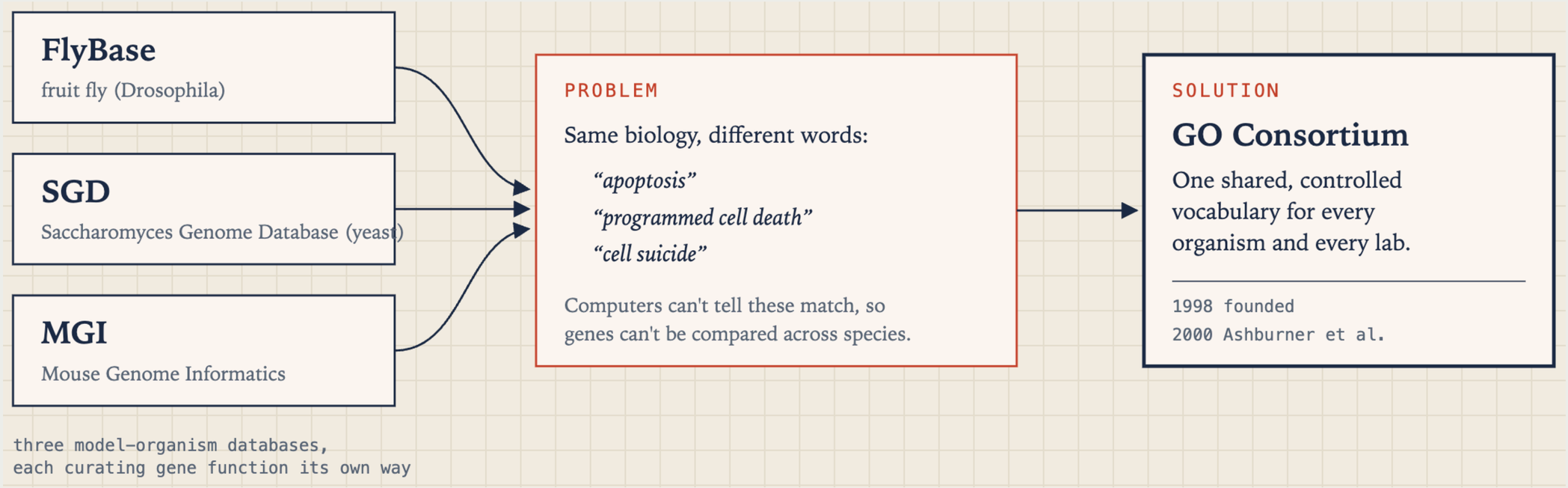
A vocabulary for describing what gene products do



Gene Ontology (GO) annotations

A vocabulary for describing what gene products do

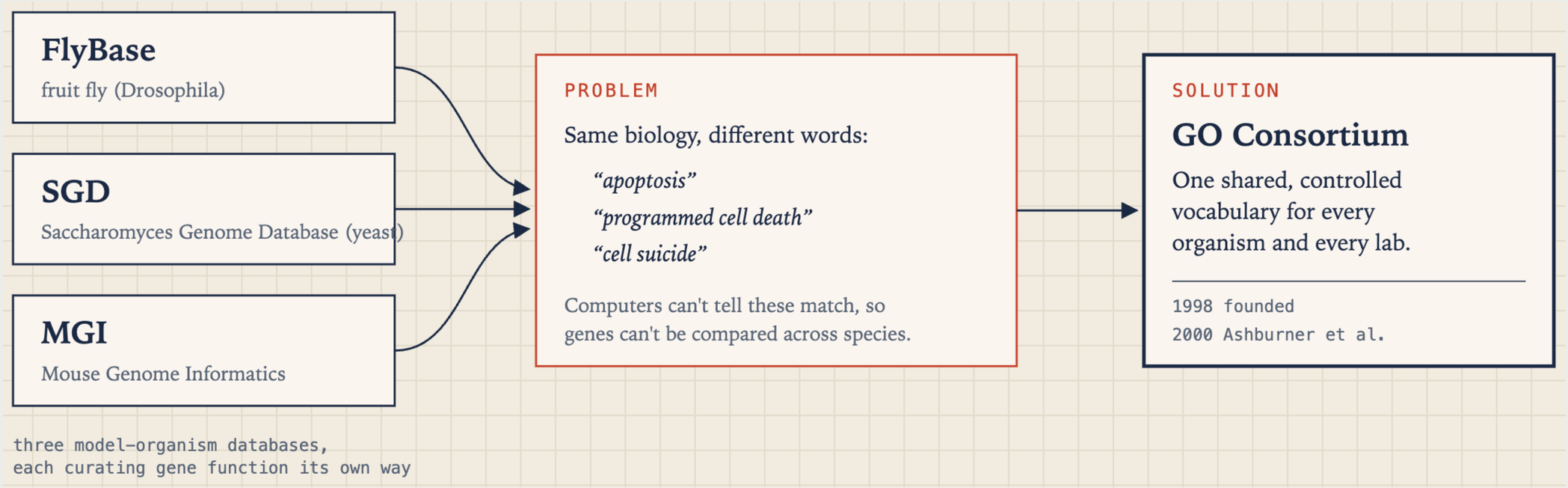
- In the 1990s, researchers needed a way to quickly link DNA sequence to the potential activity and function of the gene product



Gene Ontology (GO) annotations

A vocabulary for describing what gene products do

- In the 1990s, researchers needed a way to quickly link DNA sequence to the potential activity and function of the gene product
- In 1998, three model organism genome databases (flies, yeast, mice) collaborated to start the Gene Ontology Consortium



Gene Ontology (GO) annotations

A vocabulary for describing what gene products do

Molecular Function

What does the molecule do?

e.g. kinase activity, DNA binding

`root GO:0003674`

Biological Process

What larger program is it part of?

e.g. DNA repair, cell cycle

`root GO:0008150`

Cellular Component

Where in the cell does it act?

e.g. nucleus, mitochondrion

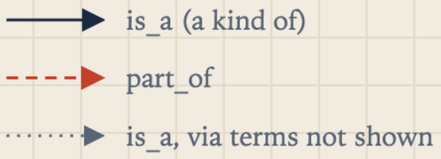
`root GO:0005575`

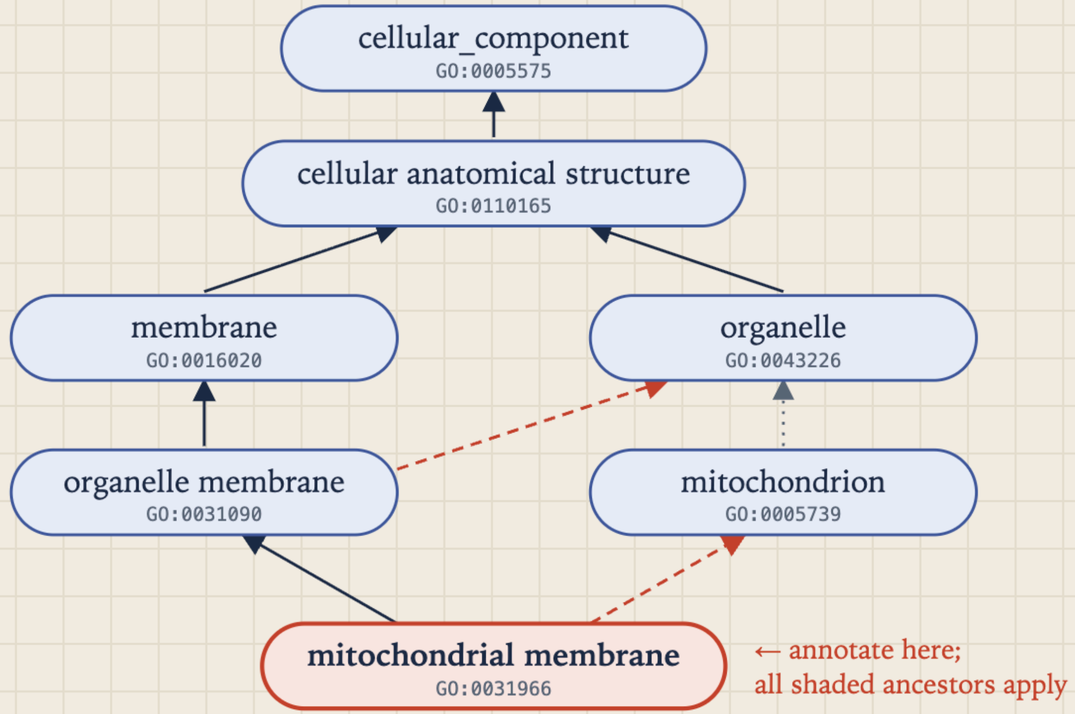
ZOOM: PART OF THE CC GRAPH

Terms run from general (top) to specific (bottom). A term can have more than one parent, so it's a graph, not a tree.

True path rule

Annotate a gene to one term and it's implicitly annotated to every ancestor, too.





Gene Ontology (GO) annotations

A vocabulary for describing what gene products do

GENE PRODUCT

TP53

human tumor suppressor protein

+

GO TERM

DNA-binding transcription factor activity

`GO:0003700 · Molecular Function`

+

EVIDENCE CODE · six categories, one example each

Experimental	**IDA**	direct assay (+ HTP variants)
Phylogenetic	**IBA**	biological aspect of ancestor
Computational	**ISS**	seq. or structural similarity
Author statement	**TAS**	traceable author statement
Curator statement	**IC**	inferred by curator
Electronic	**IEA**	not manually reviewed

+

REFERENCE

The paper or method behind the claim

↓ multiply by millions of annotations → enables

Cross-species comparison

same term means the same thing everywhere

Enrichment analysis

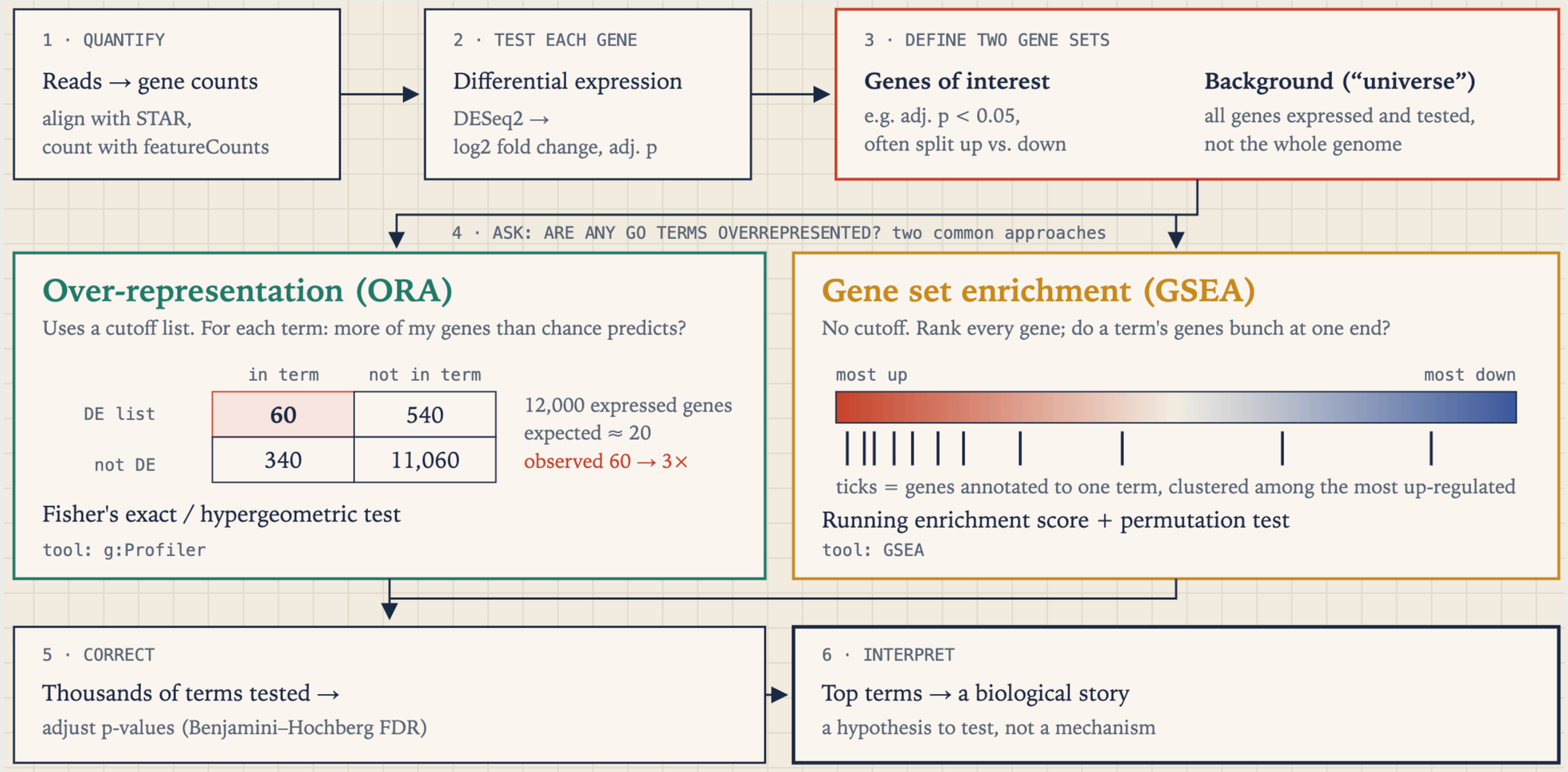
which functions are overrepresented in a list?

Weighing the evidence

experiment vs. inference, per claim

Over-representation and gene set enrichment

Using statistics to identify GO terms that are enriched/unenriched in lists of DEGS



https://zfin.org/ZDB-GENE-061103-301